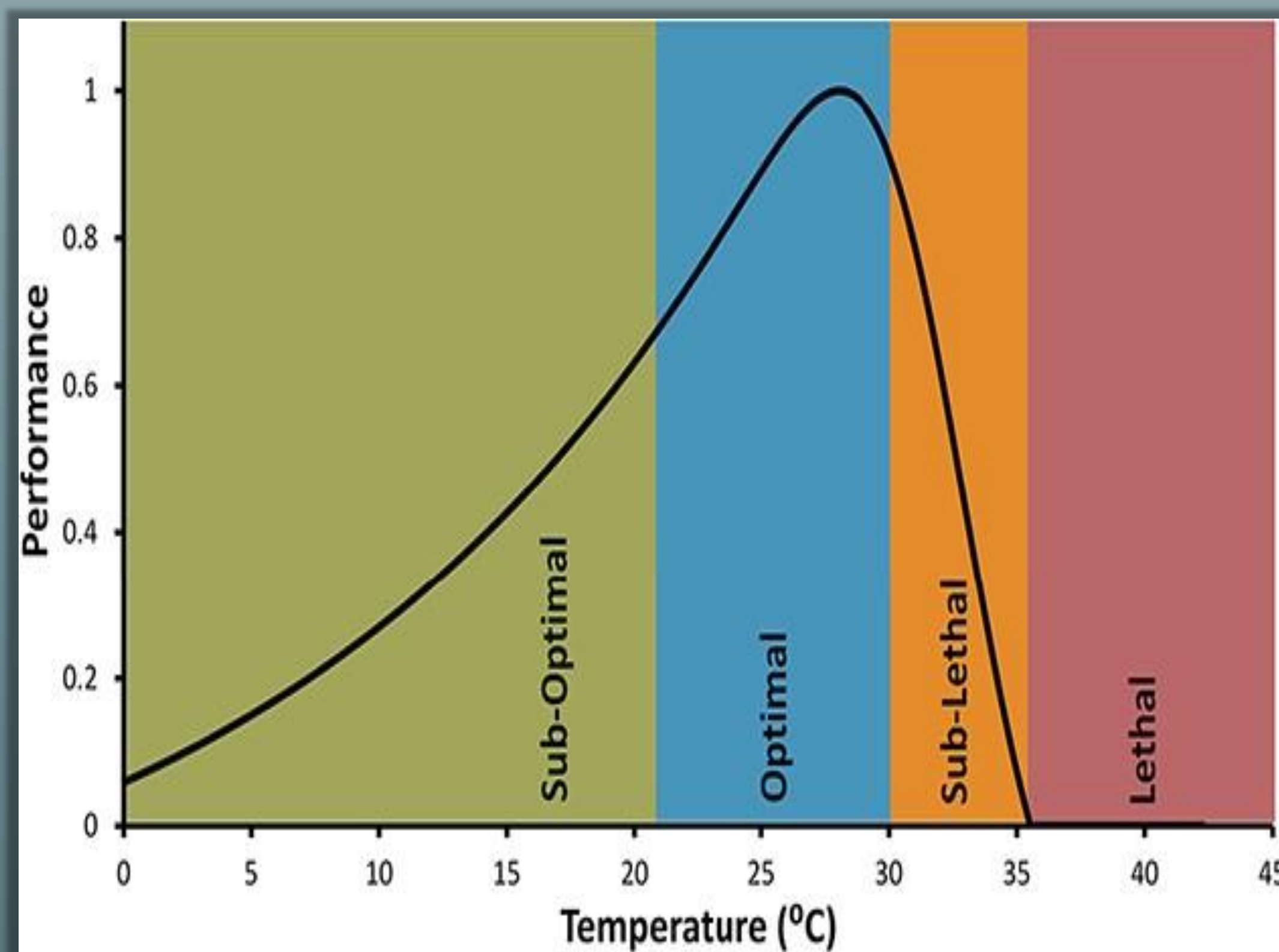


## Introduction

- Anthropogenic activities have resulted in climbing sea surface temperatures which may negatively impact marine biology<sup>4</sup>
- Parasites are integral pieces of ecology that have been found to respond negatively to increasing temperatures<sup>5</sup>
- Puget Sound exhibits a poleward temperature gradient<sup>3</sup>

## Question and Hypothesis

Do parasite populations migrate poleward as Puget Sound's temperature gradient grows stronger over time? Based on research from other Puget Sound-occupying flora and fauna<sup>1</sup>, it is likely that parasites will exhibit this pattern as the southern parts of Sound begin to breach the upper limits of the parasites' optimal temperatures<sup>2</sup>.



Choi F, Gouhier T, Lima F, Rilov G, Seabra R, Helmuth B. 2019. Mapping physiological mechanisms define scales of climate change impacts. *Conserv Physiol* 7(1): co2028.

References: (1) Berry HD, Mumford TF, Christiaan B, Dowty P, Calloway M, Ferrier L, al. 2021. Long-term changes in kelp forests in an inner basin of the Salish Sea. *PLoS ONE* 16(2): e0229703. (2) Choi F, Gouhier T, Lima F, Rilov G, Seabra R, Helmuth B. 2019. Mapping physiological mechanisms define scales of climate change impacts. *Conserv Physiol* 7(1): co2028. (3) Roethler M. 2022. Effects of Ocean Acidification and Warming on Salish Sea Kelps: a Meta-analysis. *Salish Sea Ecosystem Conference* 245. (4) Root TL, MacMynowski DP, Mastrandrea MD, Schneider SH. 2005. Human-modified temperatures induce species changes: Joint attribution. *Proc. Natl.Acad. Sci. U.S.A.* 102 (21): 7465-7469. (5) Wood CL, Velicky RL, Preisser WC, Leslie KL, Mastick N, Greene C. 2023. A reconstruction of parasite burden reveals one century of climate-associated parasite decline. *Proc. Natl.Acad. Sci. U.S.A.* 120(3): e2211903120.

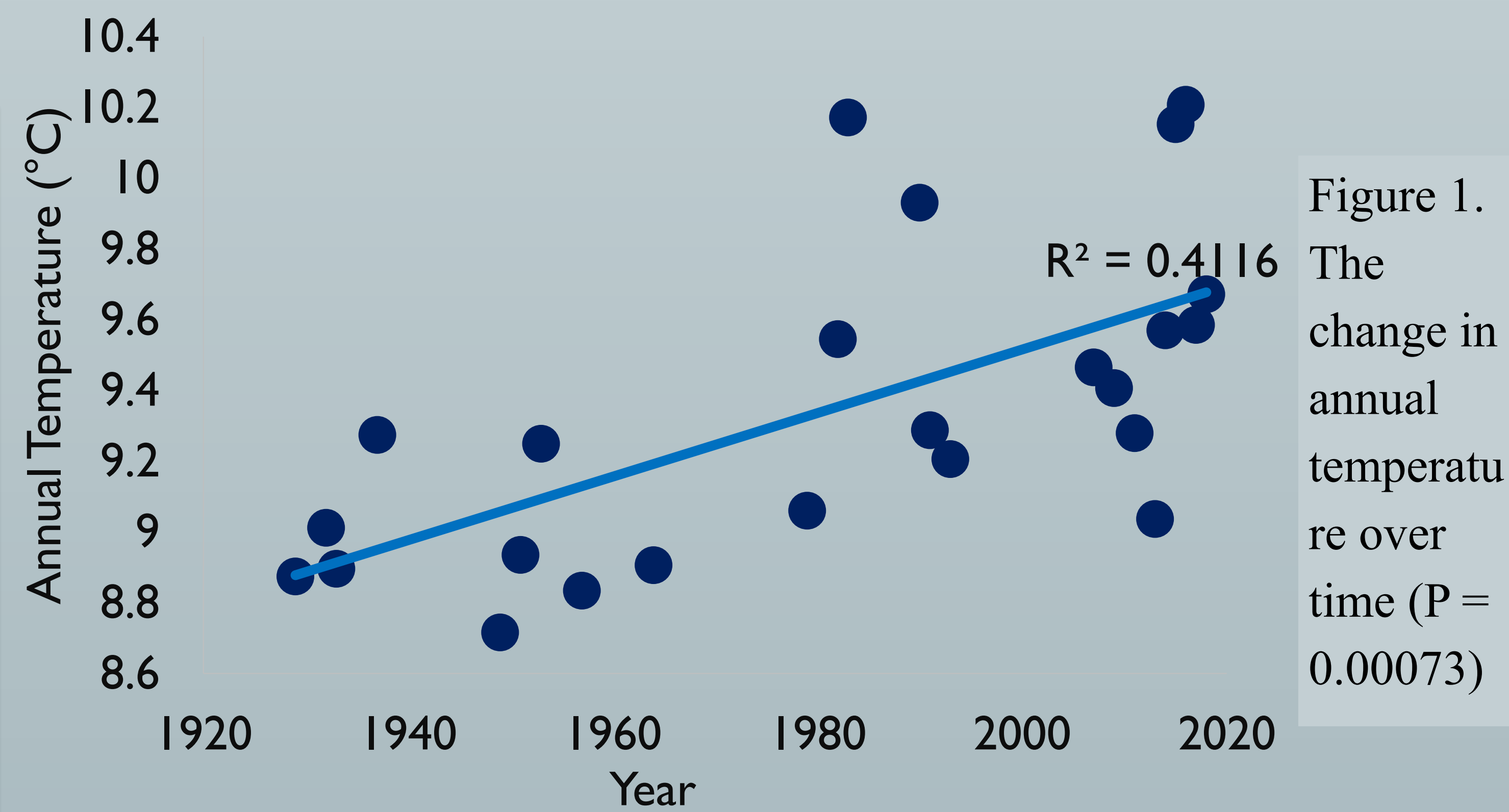


Figure 1. The change in annual temperature over time ( $P = 0.00073$ )

## Methods

- Two test methods: observing parasite migration within one host population and within all provided hosts
- Dataset from parasitological dissections of specimens in the Burke Museum's Fish Collection at the University of Washington<sup>5</sup>
- Kept relevant variables in dataset including year of catch, latitude, fish species, parasite type, parasite count, and average temperature of Puget Sound per year.

## Results

Within one fish species (pollock):

- Trematodes migrated  $0.0047^{\circ}\text{N}/\text{year}$ ,  $p = 0.096$
- Nematodes migrated  $0.0064^{\circ}\text{N}/\text{year}$ ,  $p = 0.073$
- Copepods migrated  $0.0058^{\circ}\text{N}/\text{year}$ ,  $p = 0.067$

Within all fish species:

- Poleward movement in all studied parasites with  $p$  values of 0.0099, 0.0073, and 0.00094, respectively.

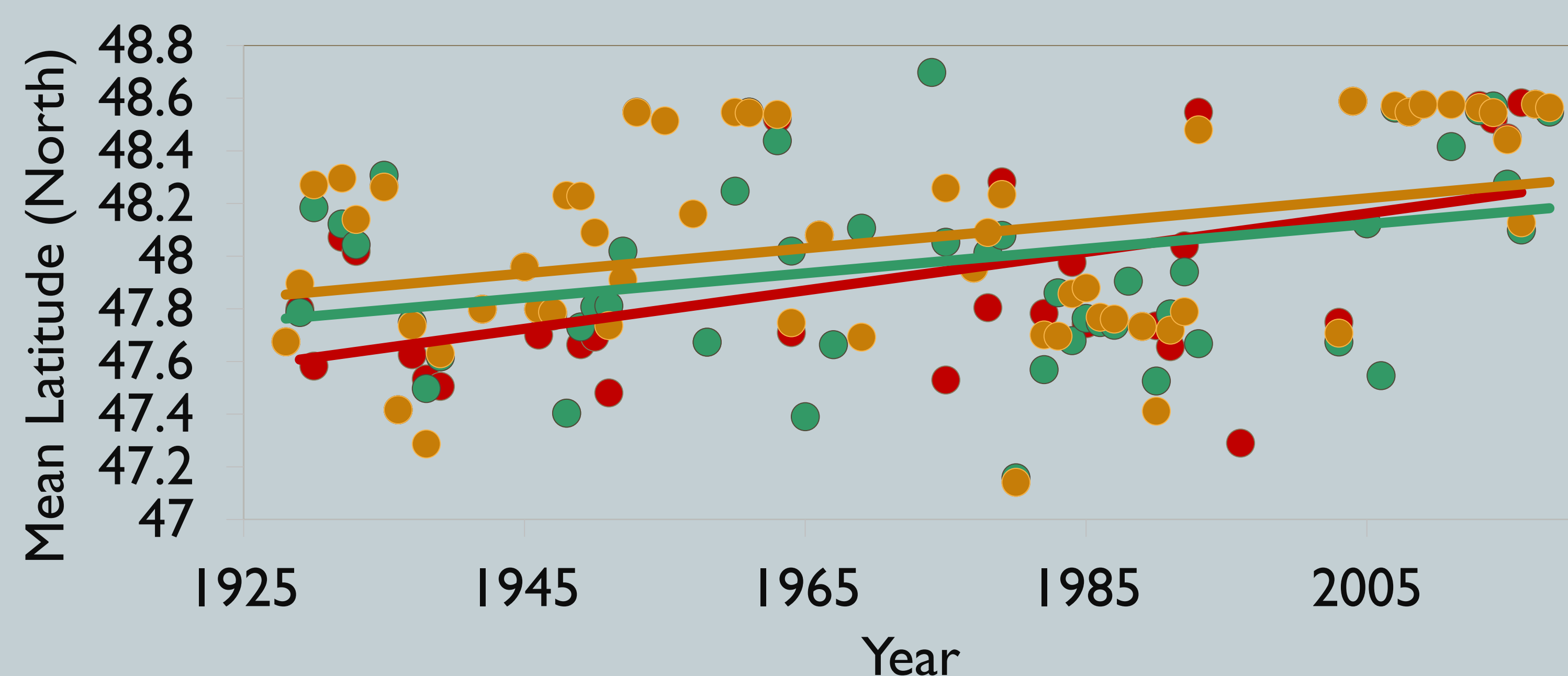


Figure 8. Mean latitude migration of copepods (red), nematodes (green), and trematodes (tan) with year among all fish species containing these parasites.

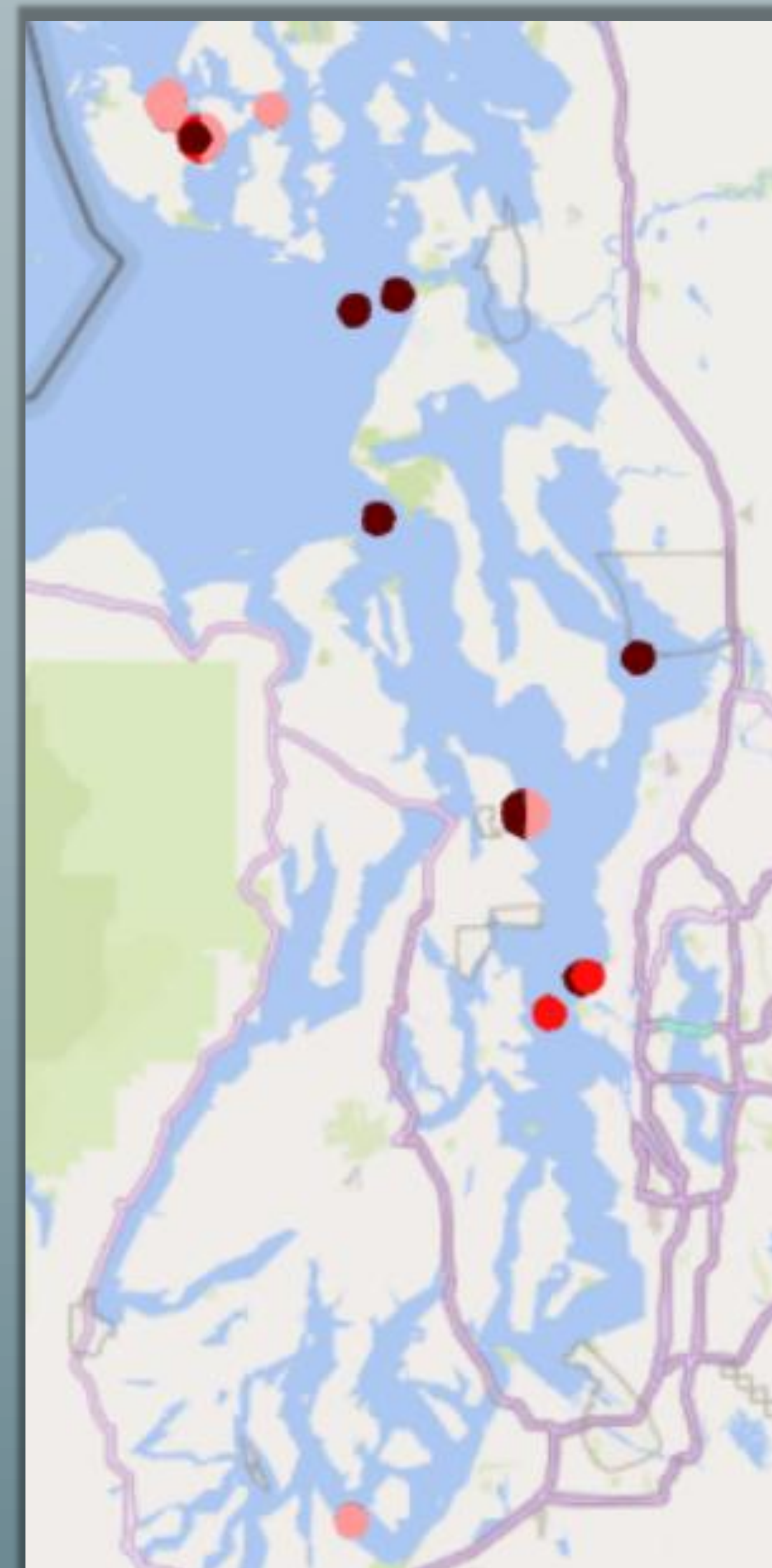


Figure 2. Spatial visualization of trematode infected fish found in Puget Sound, WA. Population was divided into thirds, each of which designated by a color of dot. Dark red dots are the oldest occurrences, and light red dots are the most modern.

Acknowledgements: Chelsea Wood, Connor Whalen, Gabby Commisso, and FISH406 students

## Discussion

Each of these parasites found within a single host species began to favor more northern latitudes. Additionally, these same types of parasites were seen to follow the same trend when all their possible hosts were analyzed, expressing an overall northern host preference.

## Conclusion

Given that some of these trends among multiple hosts were significant, continued monitoring must be conducted to record the potential continued changes in this finding into the future.